

Exploring equity standards for transport policy: A case study of calculated and perceived accessibility in the Greater Toronto and Hamilton Area, Canada

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Abstract

The lack of standards for the equitable distribution of accessibility complicates efforts to address transport poverty. This study identifies equity standards by exploring the relationship between calculated and perceived job accessibility for individuals living in the Greater Toronto and Hamilton Area, Canada. We use calculated accessibility indicators for three modes of travel (active transportation, public transit, and motorized modes) along individual-level variables to estimate ordinal models. To control for endogeneity we use spatial filters as instrumental variables. Two variables measure the perceptions of accessibility in the sample: the perception of having too few jobs available within an ideal commuting time and the perception of having to decline jobs because of transportation constraints. The results indicate that higher levels of active and transit accessibility tend to mitigate negative perceptions regarding these situations. Based on the results of these models, we identify accessibility thresholds that reduce the likelihood of negative perceptions for equity-deserving groups. We present three equity standards that we classify as conservative, intermediate, and progressive. We believe that these standards can serve as a data-driven example of how to support transportation planning in achieving more equitable outcomes, as well as for participatory processes to collectively define minimum or sufficient accessibility levels.

1. Supplementary material

Table 1 presents the total number of people living in each priority level for policy intervention, by equity standard type and census subdivision. Priority areas for policy intervention are determined with reference to the transit accessibility shortfall and to the Accessibility Fairness Index based on conservative, intermediate, and progressive equity standards.

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Table 1: People living in priority areas for policy-intervention by equity standard and census subdivision.

Subdivision	Equity standard	Priority class	Population		Census Tract	
			Total	(%)	Total	(%)
Ajax	Conservative	Very-high priority	33955	26.81	4	14.29
		High priority	41403	32.69	8	28.57
		Moderate priority	39462	31.15	9	32.14
		Low priority	11846	9.35	7	25.00
	Intermediate	Very-high priority	53400	42.16	7	25.00
		High priority	53500	42.24	11	39.29
		Moderate priority	10624	8.39	4	14.29
		Low priority	9142	7.22	6	21.43
	Progressive	Very-high priority	77869	61.48	11	39.29
		High priority	29031	22.92	7	25.00
		Moderate priority	10624	8.39	4	14.29
		Low priority	9142	7.22	6	21.43
Aurora	Conservative	Very-high priority	49696	80.08	6	66.67
		High priority	12361	19.92	3	33.33
	Intermediate	Very-high priority	49696	80.08	6	66.67
		High priority	12361	19.92	3	33.33
	Progressive	Very-high priority	49696	80.08	6	66.67
		High priority	12361	19.92	3	33.33
Brampton	Conservative	Very-high priority	217253	33.09	29	23.77
		High priority	223284	34.01	41	33.61
		Moderate priority	143894	21.92	35	28.69
		Low priority	50774	7.73	13	10.66
		No deficit	21275	3.24	4	3.28
	Intermediate	Very-high priority	348251	53.05	50	40.98
		High priority	208428	31.75	44	36.07
		Moderate priority	88734	13.52	25	20.49
		Low priority	11067	1.69	3	2.46
	Progressive	Very-high priority	380289	57.93	55	45.08
		High priority	186484	28.41	40	32.79
		Moderate priority	83110	12.66	24	19.67
		Low priority	6597	1.00	3	2.46
Burlington	Conservative	Very-high priority	4014	2.15	1	2.13
		High priority	48317	25.85	10	21.28
		Moderate priority	98680	52.78	23	48.94
		Low priority	35937	19.22	13	27.66
	Intermediate	Very-high priority	28304	15.14	5	10.64
		High priority	96536	51.64	21	44.68
		Moderate priority	54437	29.12	17	36.17
		Low priority	7671	4.10	4	8.51
	Progressive	Very-high priority	31051	16.61	5	10.64
		High priority	90968	48.66	20	42.55
		Moderate priority	52166	27.90	16	34.04
		Low priority	12763	6.83	6	12.77

Table 1: People living in priority areas for policy-intervention by equity standard and census subdivision. (*continued*)

Subdivision	Equity standard	Priority class	Population		Census Tract	
			Total	(%)	Total	(%)
Caledon	Conservative	Very-high priority	73549	96.04	12	92.31
		High priority	3032	3.96	1	7.69
	Intermediate	Very-high priority	73549	96.04	12	92.31
		High priority	3032	3.96	1	7.69
	Progressive	Very-high priority	69529	90.79	11	84.62
		High priority	7052	9.21	2	15.38
Chippewas of Georgina Island First Nation	Conservative	Low priority	231	100.00	1	100.00
	Intermediate	Low priority	231	100.00	1	100.00
	Progressive	Low priority	231	100.00	1	100.00
Clarington	Conservative	Very-high priority	70116	69.13	12	50.00
		High priority	24311	23.97	7	29.17
		Moderate priority	4695	4.63	2	8.33
		Low priority	2305	2.27	3	12.50
	Intermediate	Very-high priority	70116	69.13	12	50.00
		High priority	24311	23.97	7	29.17
		Moderate priority	4695	4.63	2	8.33
		Low priority	2305	2.27	3	12.50
	Progressive	Very-high priority	66113	65.18	11	45.83
		High priority	25710	25.35	7	29.17
		Moderate priority	7299	7.20	3	12.50
		Low priority	2305	2.27	3	12.50
East Gwillimbury	Conservative	Very-high priority	34637	100.00	4	100.00
	Intermediate	Very-high priority	34637	100.00	4	100.00
	Progressive	Very-high priority	30639	88.46	3	75.00
		High priority	3998	11.54	1	25.00
Georgina	Conservative	Very-high priority	35426	74.36	5	55.56
		High priority	12216	25.64	4	44.44
	Intermediate	Very-high priority	35426	74.36	5	55.56
		High priority	9850	20.68	3	33.33
		Moderate priority	2366	4.97	1	11.11
	Progressive	Very-high priority	35426	74.36	5	55.56
		High priority	9850	20.68	3	33.33
		Moderate priority	2366	4.97	1	11.11
Halton Hills	Conservative	Very-high priority	44671	70.96	8	61.54
		High priority	18280	29.04	5	38.46
	Intermediate	Very-high priority	44671	70.96	8	61.54
		High priority	18280	29.04	5	38.46
	Progressive	Very-high priority	44671	70.96	8	61.54
		High priority	18280	29.04	5	38.46

Table 1: People living in priority areas for policy-intervention by equity standard and census subdivision. (*continued*)

Subdivision	Equity standard	Priority class	Population		Census Tract	
			Total	(%)	Total	(%)
Hamilton	Conservative	Very-high priority	184991	32.49	32	21.48
		High priority	192625	33.84	46	30.87
		Moderate priority	156112	27.42	52	34.90
		Low priority	35572	6.25	19	12.75
	Intermediate	Very-high priority	239008	41.98	42	28.19
		High priority	189813	33.34	49	32.89
		Moderate priority	119986	21.08	44	29.53
		Low priority	20493	3.60	14	9.40
	Progressive	Very-high priority	229549	40.32	39	26.17
		High priority	166562	29.26	42	28.19
		Moderate priority	138348	24.30	47	31.54
		Low priority	34841	6.12	21	14.09
King	Conservative	Very-high priority	23830	87.18	3	75.00
		High priority	3503	12.82	1	25.00
	Intermediate	Very-high priority	23830	87.18	3	75.00
		High priority	3503	12.82	1	25.00
	Progressive	Very-high priority	23830	87.18	3	75.00
		High priority	3503	12.82	1	25.00
Markham	Conservative	Very-high priority	142066	41.97	16	26.23
		High priority	32377	9.56	7	11.48
		Moderate priority	34511	10.20	7	11.48
		Low priority	51731	15.28	13	21.31
		No deficit	77818	22.99	18	29.51
	Intermediate	Very-high priority	169165	49.97	20	32.79
		High priority	48247	14.25	10	16.39
		Moderate priority	56550	16.71	14	22.95
		Low priority	54822	16.20	15	24.59
		No deficit	9719	2.87	2	3.28
	Progressive	Very-high priority	180147	53.22	22	36.07
		High priority	53080	15.68	10	16.39
		Moderate priority	69638	20.57	18	29.51
		Low priority	35638	10.53	11	18.03
Milton	Conservative	Very-high priority	70794	53.24	12	44.44
		High priority	57490	43.23	13	48.15
		Moderate priority	4695	3.53	2	7.41
	Intermediate	Very-high priority	96956	72.91	17	62.96
		High priority	32147	24.17	8	29.63
		Moderate priority	2419	1.82	1	3.70
		Low priority	1457	1.10	1	3.70
	Progressive	Very-high priority	92863	69.83	16	59.26
		High priority	33002	24.82	8	29.63
		Moderate priority	5657	4.25	2	7.41
		Low priority	1457	1.10	1	3.70

Table 1: People living in priority areas for policy-intervention by equity standard and census subdivision. (*continued*)

Subdivision	Equity standard	Priority class	Population		Census Tract	
			Total	(%)	Total	(%)
Mississauga	Conservative	Very-high priority	50876	7.09	9	6.12
		High priority	147555	20.55	28	19.05
		Moderate priority	164253	22.88	33	22.45
		Low priority	189326	26.37	43	29.25
		No deficit	165951	23.11	34	23.13
	Intermediate	Very-high priority	133811	18.64	22	14.97
		High priority	282473	39.34	53	36.05
		Moderate priority	237360	33.06	54	36.73
		Low priority	64317	8.96	18	12.24
	Progressive	Very-high priority	203977	28.41	32	21.77
		High priority	280560	39.08	55	37.41
		Moderate priority	192896	26.87	47	31.97
		Low priority	40528	5.64	13	8.84
Newmarket	Conservative	Very-high priority	74332	84.52	12	75.00
		High priority	11373	12.93	3	18.75
		Moderate priority	2237	2.54	1	6.25
	Intermediate	Very-high priority	79356	90.24	13	81.25
		High priority	6349	7.22	2	12.50
		Moderate priority	2237	2.54	1	6.25
	Progressive	Very-high priority	79356	90.24	13	81.25
		Moderate priority	8586	9.76	3	18.75
Oakville	Conservative	Very-high priority	71533	33.47	9	21.43
		High priority	58958	27.59	12	28.57
		Moderate priority	44583	20.86	12	28.57
		Low priority	21995	10.29	6	14.29
		No deficit	16639	7.79	3	7.14
	Intermediate	Very-high priority	95120	44.51	13	30.95
		High priority	65352	30.58	15	35.71
		Moderate priority	45268	21.18	11	26.19
		Low priority	7968	3.73	3	7.14
	Progressive	Very-high priority	96846	45.32	13	30.95
		High priority	82492	38.60	18	42.86
		Moderate priority	28876	13.51	8	19.05
		Low priority	5494	2.57	3	7.14

Table 1: People living in priority areas for policy-intervention by equity standard and census subdivision. (*continued*)

Subdivision	Equity standard	Priority class	Population		Census Tract	
			Total	(%)	Total	(%)
Oshawa	Conservative	Very-high priority	73280	41.78	11	28.95
		High priority	53689	30.61	12	31.58
		Moderate priority	46619	26.58	14	36.84
		Low priority	1795	1.02	1	2.63
	Intermediate	Very-high priority	84093	47.95	13	34.21
		High priority	69283	39.50	17	44.74
		Moderate priority	20212	11.52	7	18.42
		Low priority	1795	1.02	1	2.63
	Progressive	Very-high priority	94581	53.93	15	39.47
		High priority	44986	25.65	11	28.95
		Moderate priority	34021	19.40	11	28.95
		Low priority	1795	1.02	1	2.63
Pickering	Conservative	Very-high priority	47358	47.75	7	36.84
		High priority	25053	25.26	6	31.58
		Moderate priority	8644	8.71	2	10.53
		Low priority	13271	13.38	3	15.79
	Intermediate	No deficit	4860	4.90	1	5.26
		Very-high priority	57654	58.13	9	47.37
		High priority	23911	24.11	5	26.32
		Moderate priority	17621	17.77	5	26.32
	Progressive	Very-high priority	62585	63.10	10	52.63
		High priority	18980	19.14	4	21.05
		Moderate priority	17621	17.77	5	26.32
Richmond Hill	Conservative	Very-high priority	76891	38.06	9	27.27
		High priority	17577	8.70	3	9.09
		Moderate priority	38987	19.30	6	18.18
		Low priority	37097	18.36	9	27.27
	Intermediate	No deficit	31470	15.58	6	18.18
		Very-high priority	112729	55.80	14	42.42
		High priority	29383	14.54	5	15.15
		Moderate priority	44599	22.08	10	30.30
	Progressive	Low priority	15311	7.58	4	12.12
		Very-high priority	132108	65.39	17	51.52
		High priority	29180	14.44	5	15.15
		Moderate priority	34686	17.17	9	27.27
		Low priority	6048	2.99	2	6.06

Table 1: People living in priority areas for policy-intervention by equity standard and census subdivision. (*continued*)

Subdivision	Equity standard	Priority class	Population		Census Tract	
			Total	(%)	Total	(%)
Toronto	Conservative	Very-high priority	29100	1.04	5	0.86
		High priority	99642	3.57	18	3.09
		Moderate priority	139501	4.99	27	4.63
		Low priority	467891	16.75	102	17.50
		No deficit	2057959	73.65	431	73.93
	Intermediate	Very-high priority	134759	4.82	20	3.43
		High priority	337150	12.07	59	10.12
		Moderate priority	637881	22.83	121	20.75
		Low priority	1074987	38.47	253	43.40
		No deficit	609316	21.81	130	22.30
	Progressive	Very-high priority	221768	7.94	32	5.49
		High priority	663650	23.75	108	18.52
		Moderate priority	807907	28.91	155	26.59
		Low priority	1095029	39.19	287	49.23
		No deficit	5739	0.21	1	0.17
Uxbridge	Conservative	Very-high priority	21556	100.00	4	100.00
	Intermediate	Very-high priority	21556	100.00	4	100.00
	Progressive	Very-high priority	17528	81.31	3	75.00
		High priority	4028	18.69	1	25.00
Vaughan	Conservative	Very-high priority	144670	44.78	17	31.48
		High priority	29157	9.02	6	11.11
		Moderate priority	23837	7.38	5	9.26
		Low priority	23930	7.41	7	12.96
		No deficit	101509	31.42	19	35.19
	Intermediate	Very-high priority	168654	52.20	21	38.89
		High priority	38634	11.96	7	12.96
		Moderate priority	51109	15.82	11	20.37
		Low priority	44381	13.74	11	20.37
		No deficit	20325	6.29	4	7.41
	Progressive	Very-high priority	178915	55.37	22	40.74
		High priority	51859	16.05	10	18.52
		Moderate priority	56645	17.53	12	22.22
		Low priority	35684	11.04	10	18.52
Whitby	Conservative	Very-high priority	52649	38.01	8	26.67
		High priority	41495	29.96	8	26.67
		Moderate priority	39135	28.26	12	40.00
		Low priority	5222	3.77	2	6.67
	Intermediate	Very-high priority	64036	46.24	10	33.33
		High priority	51701	37.33	11	36.67
		Moderate priority	22764	16.44	9	30.00
		Very-high priority	75070	54.20	12	40.00
	Progressive	High priority	40667	29.36	9	30.00
		Moderate priority	17167	12.39	6	20.00
		Low priority	5597	4.04	3	10.00

Table 1: People living in priority areas for policy-intervention by equity standard and census subdivision. (*continued*)

Subdivision	Equity standard	Priority class	Population		Census Tract	
			Total	(%)	Total	(%)
Whitchurch-Stouffville	Conservative	Very-high priority	49864	100.00	7	100.00
	Intermediate	Very-high priority	49864	100.00	7	100.00
	Progressive	Very-high priority	49864	100.00	7	100.00